

04:00

NIGHTTRACE

COMBAT SYSTEMS CODEX

Weapons · Evolutions · Bosses · Difficulty · Mobile Readiness



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WEAPON LINES

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PAIRED MODULES

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LEVELS

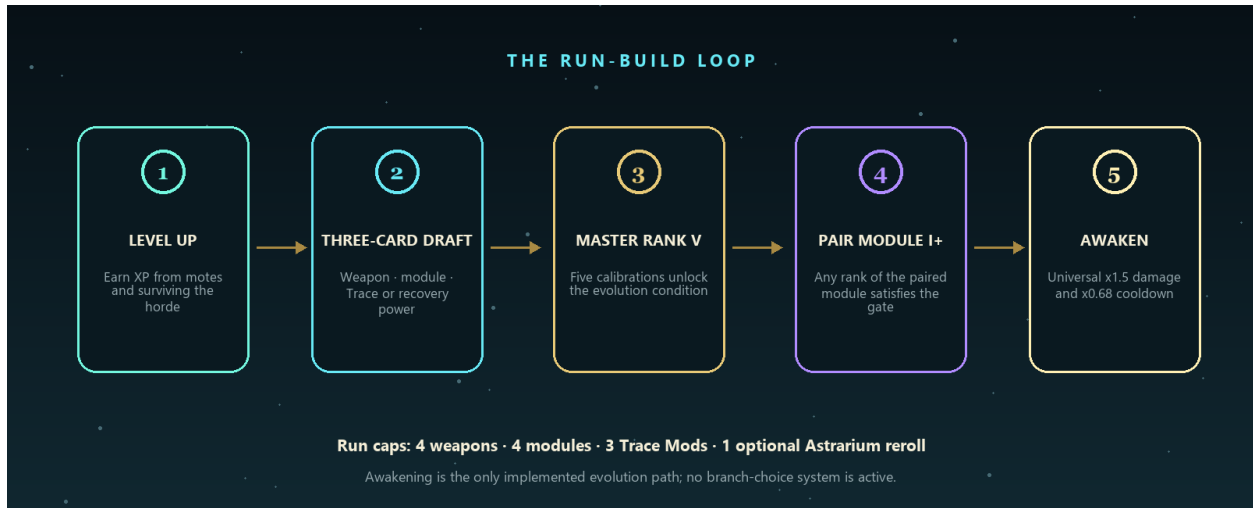
RUNTIME-CAPTURED REFERENCE · v1.6.0 · 2026.07.25

Prepared from the current NIGHTTRACE implementation

FIELD GUIDE 00

How to Read This Codex

A player-facing reference and an implementation-accurate audit of the current build.



CURRENT CONTENT AT A GLANCE

8 weapons · 8 paired modules · 8 Trace Mods · 3 recovery/overdrive drafts · 10 permanent Astrarium nodes · 10 bosses · 10 levels.

State language

- Solo means Rank I with no module. Combined means Rank I plus the paired Module I.
- Mastered means Rank V with no module. Final means Awakened Rank V plus the paired Module III.
- Awakening requires Rank V plus at least Rank I of the paired module. The module is not consumed.
- Every awakening applies x1.5 core damage and x0.68 cooldown. Graveglass Spires and Eclipse Harrow expand their remote formations while preserving one damage application per target, even where visual strikes overlap.
- Every weapon plate is composed from four deterministic 1440 x 900 frames produced by the live Pixi gameplay renderer, frozen at a weapon-specific authored readability beat.

CAPTURE PROTOCOL

One exact weapon loadout, eight deterministic target dummies, fixed local-only showcase route, production fire logic, and a fixed post-fire sample time. The 32 source PNGs and their SHA-256 hashes are listed in capture-manifest.json.

Sources: capture-manifest.json · content.ts · GameCanvas.tsx · showcase.ts

FIELD GUIDE 01

Combat Presentation Rationale

v1.6.0 gives every major event its own origin, path, target, and decay grammar instead of making every effect radiate from the bearer.

System	Spatial / motion rule	Runtime result
Graveglass Spires	Choose the densest remote cluster; warn through floor fractures; erupt staggered nightglass; shed fragments quickly.	World-anchored branch formations scale 2 / 3 / 4 / 6 across Solo, Combined, Mastered, and Final.
Eclipse Harrow	Predict moving lanes; stage execution planes off the player; cut decisively; collapse before the next threat cue.	Single, crossed, staggered, then cathedral formations scale 1 / 2 / 3 / 4 without repeated overlap damage.
Rare support relics	A persistent tapered shaft descends from above, lands on a broken rune, and remains legible through horde density.	Dawnheart, Gravestar, and Pulse Core keep their timed utility while gaining a strong heaven-to-ground pickup landmark.
Sovereign attacks	Identity comes from geometry, telegraph direction, charge lanes, and gather-release-aftershock body choreography.	All ten bosses retain distinct idle, locomotion, melee, and level-specific special silhouettes.
Music	Ambient dungeon tension yields to a denser original boss arrangement through lifecycle-safe crossfades.	Original NIGHTTRACE music is separate from reference audio; boss entry, phase escalation, victory, and defeat remain distinct events.

INTERNAL REFERENCE ANALYSIS - NO BORROWED ASSETS OR AUDIO

A 442.19-second, 560 x 1312 gameplay recording at approximately 57.8 fps was inspected for pacing and hierarchy. The observed combat pulse was about 136 BPM (68 half-time); reference loudness measured -26.5 LUFS integrated, 10.3 LU LRA, and -4.3 dBFS true peak, while gameplay spectral centroid sat around 2.3-2.9 kHz versus roughly 1.1 kHz in menus. These measurements informed contrast and pacing only; NIGHTTRACE uses its own code, artwork, effects, and music.

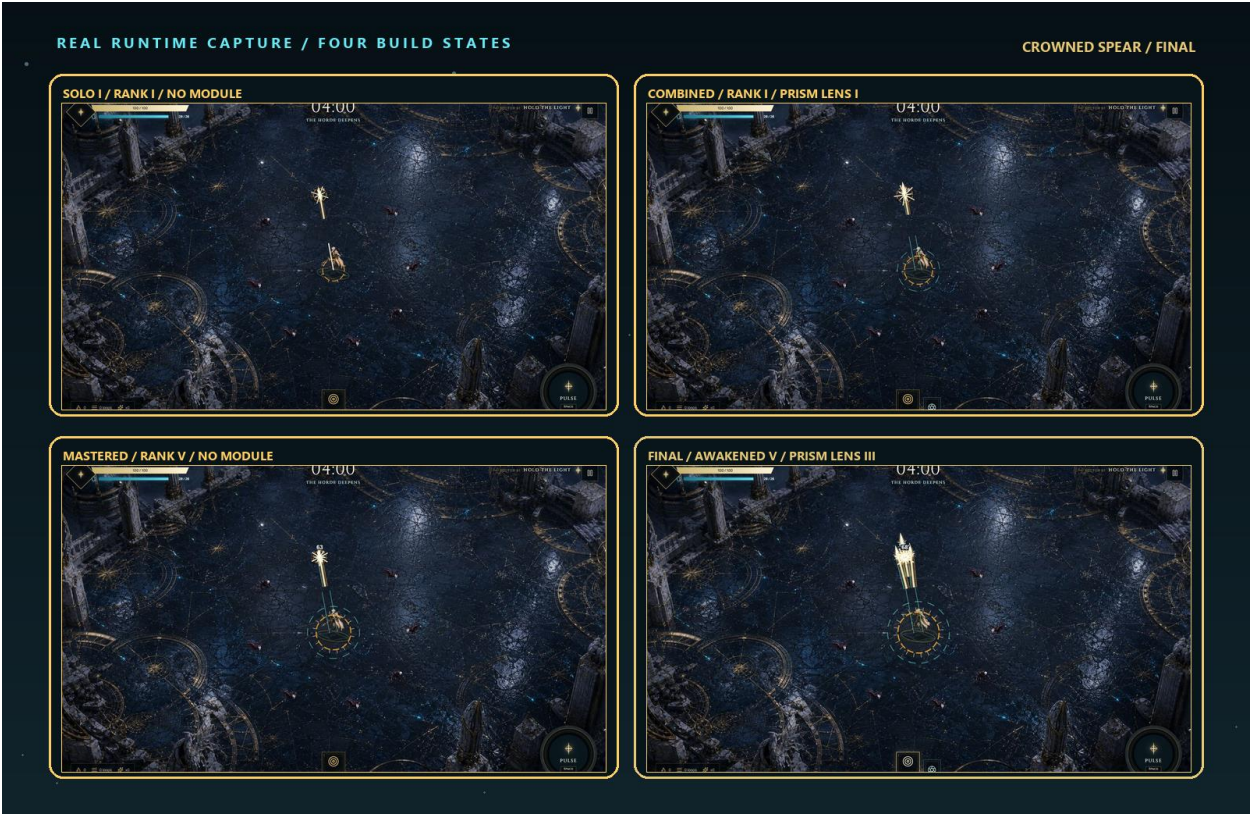
- Anticipation must reveal where danger or power will land before the decisive impact.
- The impact owns the strongest contrast; decay is rapid unless persistence communicates an uncollected reward.
- Not every allied power originates from the bearer. Remote world anchors create tactical legibility and a larger sense of place.
- Boss prominence is earned through special geometry and motion authority, while ordinary hordes stay subordinate.

Sources: internal reference-video frame/audio analysis · weaponPatterns.ts · weaponVfx.ts · pickupPresentation.ts · animation.ts · audio.ts

WEAPON 01 / 08

Helio Lance

A solar needle that spears the nearest target. Paired module: Prism Lens.



Actual gameplay frames: Solo I, Combined I + Module I, Mastered V, and Final Awakened V + Module III.

State	Core damage	Base cooldown	Pattern / effective change
Solo I	28.00	0.580s	1 needle · pierce 1 · speed 720
Combined I + Module I	30.24	0.560s	1 needle · +8% damage · speed 775 · +0.14s life
Mastered V	62.72	0.452s	1 needle · pierce 2 · speed 720
Final V + Module III	116.66	0.266s	Main + 2 auxiliary needles · module damage/speed/life III

CROWNED SPEAR

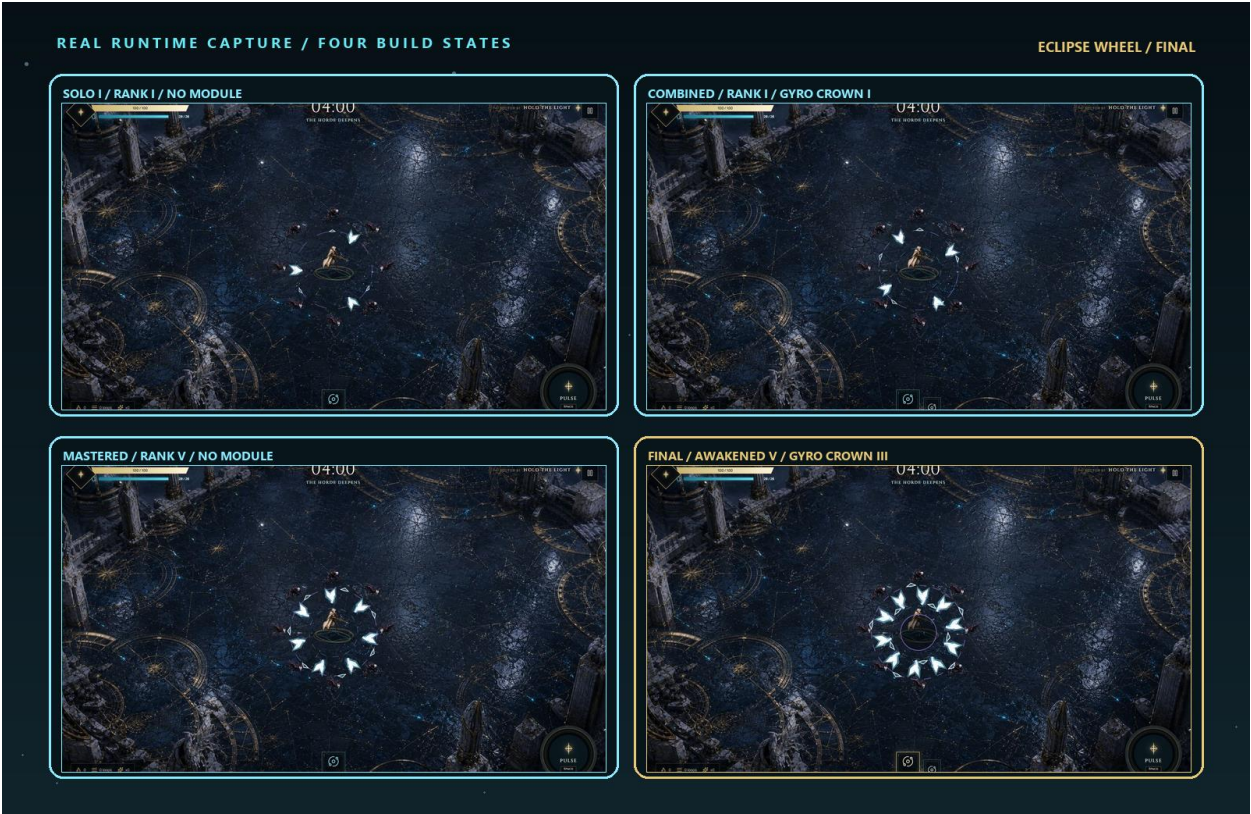
Each rank: +8% raw damage, +55 speed, +0.14s life; -3.5% cooldown. The awakening adds a distinct offensive pattern.

Sources: helio-lance/solo.png, combined.png, mastered.png, final.png · capture-manifest.json · content.ts · GameCanvas.tsx

WEAPON 02 / 08

Crescent Array

A radial fan of moonblades carves space around the bearer. Paired module: Gyro Crown.



Actual gameplay frames: Solo I, Combined I + Module I, Mastered V, and Final Awakened V + Module III.

State	Core damage	Base cooldown	Pattern / effective change
Solo I	18.00	1.050s	3 radial blades
Combined I + Module I	18.00	1.013s	4 radial blades
Mastered V	40.32	0.819s	7 radial blades
Final V + Module III	60.48	0.482s	10 radial blades (cap)

ECLIPSE WHEEL

Each rank: +1 blade; -3.5% cooldown. Runtime orbit speed/pierce remain fixed. The awakening adds a distinct offensive pattern.

Sources: crescent-array/solo.png, combined.png, mastered.png, final.png · capture-manifest.json · content.ts · GameCanvas.tsx

WEAPON 03 / 08

Arc Choir

Resonant lightning jumps between enemies in a rising chord. Paired module: Resonance Coil.



Actual gameplay frames: Solo I, Combined I + Module I, Mastered V, and Final Awakened V + Module III.

State	Core damage	Base cooldown	Pattern / effective change
Solo I	24.00	1.280s	3-target chain
Combined I + Module I	25.68	1.235s	4-target chain · +7% initial damage
Mastered V	53.76	0.998s	7-target chain
Final V + Module III	97.57	0.588s	10-target chain (cap) · +21% initial damage

CATHEDRAL STORM

Each rank: +1 target and +7% initial raw damage; -3.5% cooldown. The awakening adds a distinct offensive pattern.

Sources: arc-choir/solo.png, combined.png, mastered.png, final.png · capture-manifest.json · content.ts · GameCanvas.tsx

WEAPON 04 / 08

Rift Seeds

Plants unstable wells that bloom into short-lived singularities. Paired module: Grav Anchor.



Actual gameplay frames: Solo I, Combined I + Module I, Mastered V, and Final Awakened V + Module III.

State	Core damage	Base cooldown	Pattern / effective change
Solo I	38.00	1.720s	1 seed · 92-radius bloom
Combined I + Module I	42.56	1.660s	1 seed · +12% damage · +0.12s life
Mastered V	85.12	1.342s	1 seed · mastered core
Final V + Module III	173.64	0.789s	Universal awakening + final satellite-ring VFX

EVENTIDE GARDEN

Each rank: +12% raw damage, +0.12s life, +12% pickup radius; -3.5% cooldown. The awakening adds universal power plus a final-state visual signature.

Sources: rift-seeds/solo.png, combined.png, mastered.png, final.png · capture-manifest.json · content.ts · GameCanvas.tsx

WEAPON 05 / 08

Comet Swarm

A hunting flight of micro-comets pursues priority targets. Paired module: Guidance Filament.



Actual gameplay frames: Solo I, Combined I + Module I, Mastered V, and Final Awakened V + Module III.

State	Core damage	Base cooldown	Pattern / effective change
Solo I	21.00	0.920s	2 guided comets
Combined I + Module I	22.26	0.888s	2 comets · +6% damage · +35 speed · +0.8 homing
Mastered V	47.04	0.718s	4 guided comets
Final V + Module III	83.26	0.422s	6 guided comets · Module III guidance

PERIHELION HUNT

Each rank: +6% raw damage, +35 speed, +0.8 homing; -3.5% cooldown. The awakening adds a distinct offensive pattern.

Sources: comet-swarm/solo.png, combined.png, mastered.png, final.png · capture-manifest.json · content.ts · GameCanvas.tsx

WEAPON 06 / 08

Graveglass Spires

Fractured nightglass erupts beneath the densest enemy cluster, away from the bearer. Paired module: Ossuary Prism.



Actual gameplay frames: Solo I, Combined I + Module I, Mastered V, and Final Awakened V + Module III.

State	Core damage	Base cooldown	Pattern / effective change
Solo I	34.00	1.080s	2 remote spires · densest-cluster anchor
Combined I + Module I	37.40	1.042s	3 remote spires · +10% damage · +14 cluster reach
Mastered V	76.16	0.842s	4 staggered spires · mastered branch
Final V + Module III	148.51	0.496s	6 spires · Sepulchral Orchard · one hit per target

SEPULCHRAL ORCHARD

Each rank: +10% raw damage and +14 cluster reach; -3.5% cooldown. The awakening adds universal power plus a final-state visual signature.

Sources: graveglass-spires/solo.png, combined.png, mastered.png, final.png · capture-manifest.json · content.ts · GameCanvas.tsx

WEAPON 07 / 08

Mirror Bow

Paired hard-light bolts pierce forward and along the mirrored heading. Paired module: Flux Mirror.



Actual gameplay frames: Solo I, Combined I + Module I, Mastered V, and Final Awakened V + Module III.

State	Core damage	Base cooldown	Pattern / effective change
Solo I	34.00	0.740s	Forward pierce 3 + reverse 75% bolt
Combined I + Module I	36.72	0.714s	Forward pierce 4 · +8%; reverse unchanged
Mastered V	76.16	0.577s	Forward pierce 5 + reverse bolt
Final V + Module III	141.66	0.340s	Forward pierce 8 · +24% + twin final echo VFX

INFINITE REFRAIN

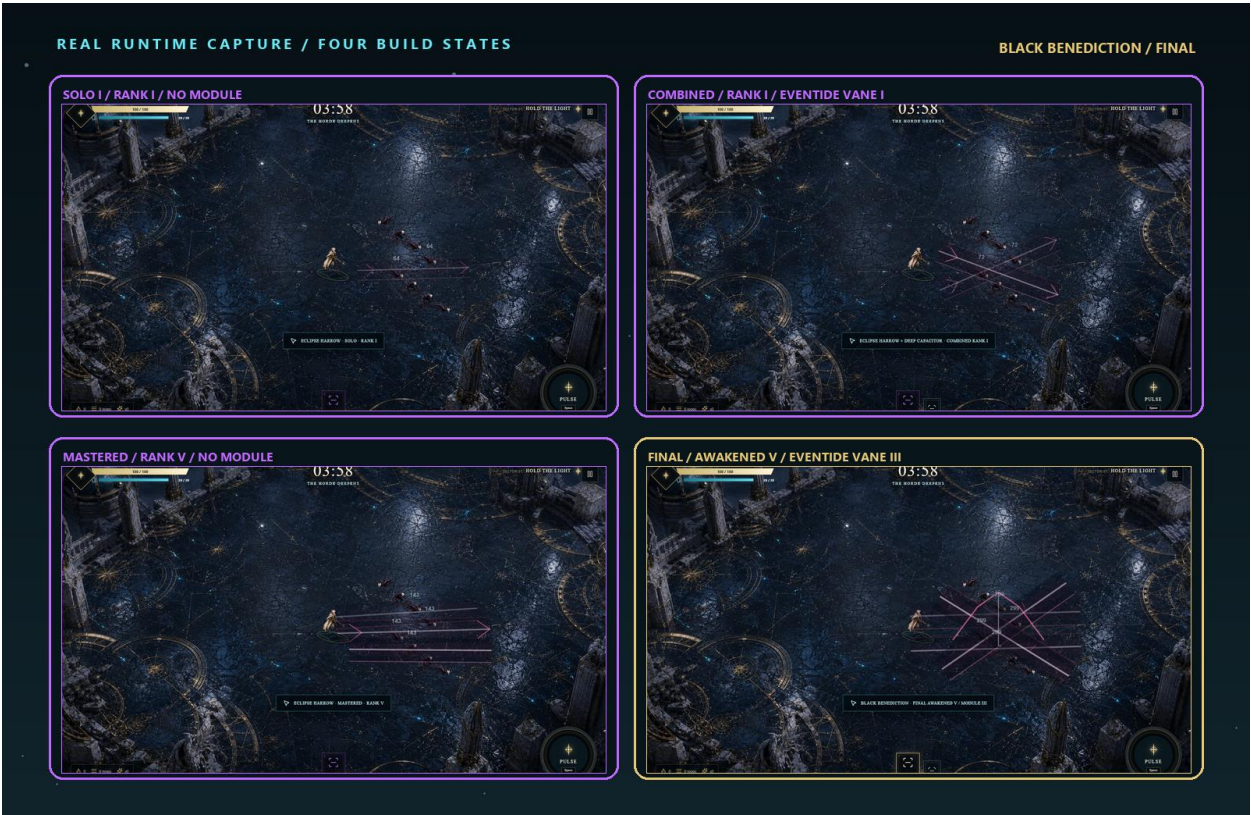
Each rank: +8% forward damage and +1 forward pierce; -3.5% cooldown. The awakening adds universal power plus a final-state visual signature.

Sources: mirror-bow/solo.png, combined.png, mastered.png, final.png · capture-manifest.json · content.ts · GameCanvas.tsx

WEAPON 08 / 08

Eclipse Harrow

Nightglass execution gates cut predicted enemy lanes at a remote world anchor. Paired module: Eventide Vane.



Actual gameplay frames: Solo I, Combined I + Module I, Mastered V, and Final Awakened V + Module III.

State	Core damage	Base cooldown	Pattern / effective change
Solo I	64.00	2.120s	1 predicted execution lane
Combined I + Module I	72.32	2.046s	2 crossed lanes · +13% damage · +0.04s prediction
Mastered V	143.36	1.654s	3 staggered mastered lanes
Final V + Module III	298.91	0.973s	4 cathedral lanes · Black Benediction · one hit per target

BLACK BENEDICTION

Each rank: +13% raw damage and +0.04s target prediction; -3.5% cooldown. The awakening adds universal power plus a final-state visual signature.

Sources: eclipse-harrow/solo.png, combined.png, mastered.png, final.png · capture-manifest.json · content.ts · GameCanvas.tsx

SUPPORT SYSTEMS 01

Paired Modules

Modules scale from Rank I to III and gate their weapon's awakening at Rank I.

PAIRED MODULES · RANK I → III ·

Every module rank also removes 3.5% from its paired weapon's base cooldown factor.



PRISM LENS

PAIRS WITH HELIO LANCE

+8% damage · +55 speed · +0.14s life

Reach is speed/lifetime; no mark system.

◆◆◆



GYRO CROWN

PAIRS WITH CRESCENT ARRAY

+1 blade

Orbit rate and projectile pierce stay fixed.

◆◆◆



RESONANCE COIL

PAIRS WITH ARC CHOIR

+1 target · +7% initial damage

Chain radius stays 320; jumps fall off.

◆◆◆



GRAV ANCHOR

PAIRS WITH RIFT SEEDS

+12% damage · +0.12s life · +12% pickup

No enemy hold; bloom radius stays 92.

◆◆◆



GUIDANCE FILAMENT

PAIRS WITH COMET SWARM

+6% damage · +35 speed · +0.8 homing

Fully implemented.

◆◆◆



OSSUARY PRISM

PAIRS WITH GRAVEGLASS SPIRES

+10% damage · +14 cluster reach

Coverage grows at the selected remote cluster.

◆◆◆



FLUX MIRROR

PAIRS WITH MIRROR BOW

+8% forward damage · +1 forward pierce

Reverse bolt is unchanged.

◆◆◆



EVENTIDE VANE

PAIRS WITH ECLIPSE HARROW

+13% damage · +0.04s prediction

Prediction leads moving targets; no repeated overlap damage.

◆◆◆

All module ranks also reduce the paired weapon's base cooldown factor by 0.035.

Module	Rank I	Rank II	Rank III	Awakening gate
Prism Lens	1× effect	2× effect	3× effect · Elite	Helio Lance V + this I+
Gyro Crown	1× effect	2× effect	3× effect · Elite	Crescent Array V + this I+
Resonance Coil	1× effect	2× effect	3× effect · Elite	Arc Choir V + this I+
Grav Anchor	1× effect	2× effect	3× effect · Elite	Rift Seeds V + this I+
Guidance Filament	1× effect	2× effect	3× effect · Elite	Comet Swarm V + this I+
Ossuary Prism	1× effect	2× effect	3× effect · Elite	Graveglass Spires V + this I+
Flux Mirror	1× effect	2× effect	3× effect · Elite	Mirror Bow V + this I+
Eventide Vane	1× effect	2× effect	3× effect · Elite	Eclipse Harrow V + this I+

COPY AND RUNTIME

The current player-facing module descriptions now name the effects that are actually live. This page adds exact rank math and boundary notes so cadence, range, count, and damage are not conflated.

Sources: content.ts MODULES / buildUpgradeOptions · GameCanvas.tsx updateWeapons / fireWeapon

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SUPPORT SYSTEMS 02

Trace Mods & Recovery Powers

Trace Mods are one-rank elites; a run can hold three distinct mods.

TRACE MODS · SINGLE-RANK ELITES



CLOSED CIRCUIT
Closed loops deal x1.35 damage.



AFTERIMAGE
+22 retained Trace points.



CROSSFIRE
Every 4th weapon proc fires one opposite 72% bolt.



NIGHTGLASS
+6% crit chance; crits deal x1.75.



FAULTLINE
Hits ≥45 fracture 18% raw damage within 86.



SUNBLOOD
Every 20th non-boss kill heals 3.5% max HP.



QUIET STEP
+18% speed after 4s without damage.



RED SHIFT
Nearby density cuts cooldown up to 28%.

RECOVERY / OVERDRIVE DRAFTS

VITAL REPAIR

RECOVERY

Restore 30% max HP.

AEGIS OVERCHARGE

RECOVERY

+6 max shield for the run; fully restore shield.

PULSE PRIME

OVERDRIVE

Set Dawn Pulse charge to 100%.

All eight unowned Trace Mods can enter drafts; Codex discovery does not gate the candidate pool.

Power	Trigger	Result
Trace Loop	Close within 30px or intersect; area ≥2,100	Base 34 + 5.5×level + area bonus; bosses receive x0.65
Dawn Pulse	Charge reaches 100 and player activates	120 + 11×level raw damage in radius 610; distance falloff
Crossfire	Every fourth global weapon proc	One opposite 72% bolt—not a copy of the full original pattern
Pulse Prime	Draft selection	Fills Pulse to 100%; invulnerability is not granted on selection

Sources: content.ts:154–195, 566–762 · GameCanvas.tsx:639–670, 1386–1530, 1780–1937

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PERMANENT PROGRESSION 01

The Astrarium

Ten persistent nodes transform starting durability, damage, mobility, drafting, and Trace mastery.



A dependency is satisfied at parent Rank I; refunding returns all historical shard costs.

Node	Cap	Incremental shard costs	Full-rank effect
Vital Spark	5	18 / 36 / 54 / 72 / 90	Max HP: 100 → 120
Aegis Thread	5	22 / 44 / 66 / 88 / 110	Shield: 28 → 43
Helio Focus	5	26 / 52 / 78 / 104 / 130	All damage: x1.00 → x1.10
Long Memory	4	30 / 60 / 90 / 120	Trace cap: 72 → 84 points
Mote Song	4	24 / 48 / 72 / 96	Pickup radius: 132 → 168.96
Bright Draft	1	70	One reroll per run
Red Shift	4	32 / 64 / 96 / 128	Move speed: 235 → 249.1
Parallax Eye	3	42 / 84 / 126	Crit: 0% → 4.5%
Echo Chamber	3	46 / 92 / 138	Loop damage: x1.00 → x1.15
Dawn Within	1	120	Boss damage: x1.08

Sources: ui/data.ts:22-155 · save.ts:43-79 · App.tsx:212-242 · GameCanvas.tsx:387-425

BOSS SYSTEMS 01

Boss Attack Pattern Atlas • Levels 1–5

Each sector now owns a distinct three-phase telegraph signature.

BOSS TELEGRAPH ATLAS • LEVELS 1-5 • PHASE I → III

01 LINE FAN

GLOAM STAG

2 / 3 / 4 aimed lines • x1.15 damage

PHASE 1

PHASE 2

PHASE 3

02 ORBIT RING

MIRE CANTOR

4 / 5 / 6 circles • +0.12s warning

PHASE 1

PHASE 2

PHASE 3

03 CROSS LANES

RAILJAW PRIME

2 / 2 / 4 lanes • x1.12 damage

PHASE 1

PHASE 2

PHASE 3

04 MIRROR MARKS

MIRROR MATRON

Current + reflected mark; aimed line P2+

PHASE 1

PHASE 2

PHASE 3

05 SPIRAL CLUSTER

TIDE APOSTLE

3 / 4 / 5 circles • 0.06s stagger

PHASE 1

PHASE 2

PHASE 3

Warnings compress from 0.82s → 0.74s → 0.66s as boss health crosses 67% and 34%.

Boss phase thresholds: Phase II at <=67% HP; Phase III at <=34% HP. Warning time: 0.82s / 0.74s / 0.66s.

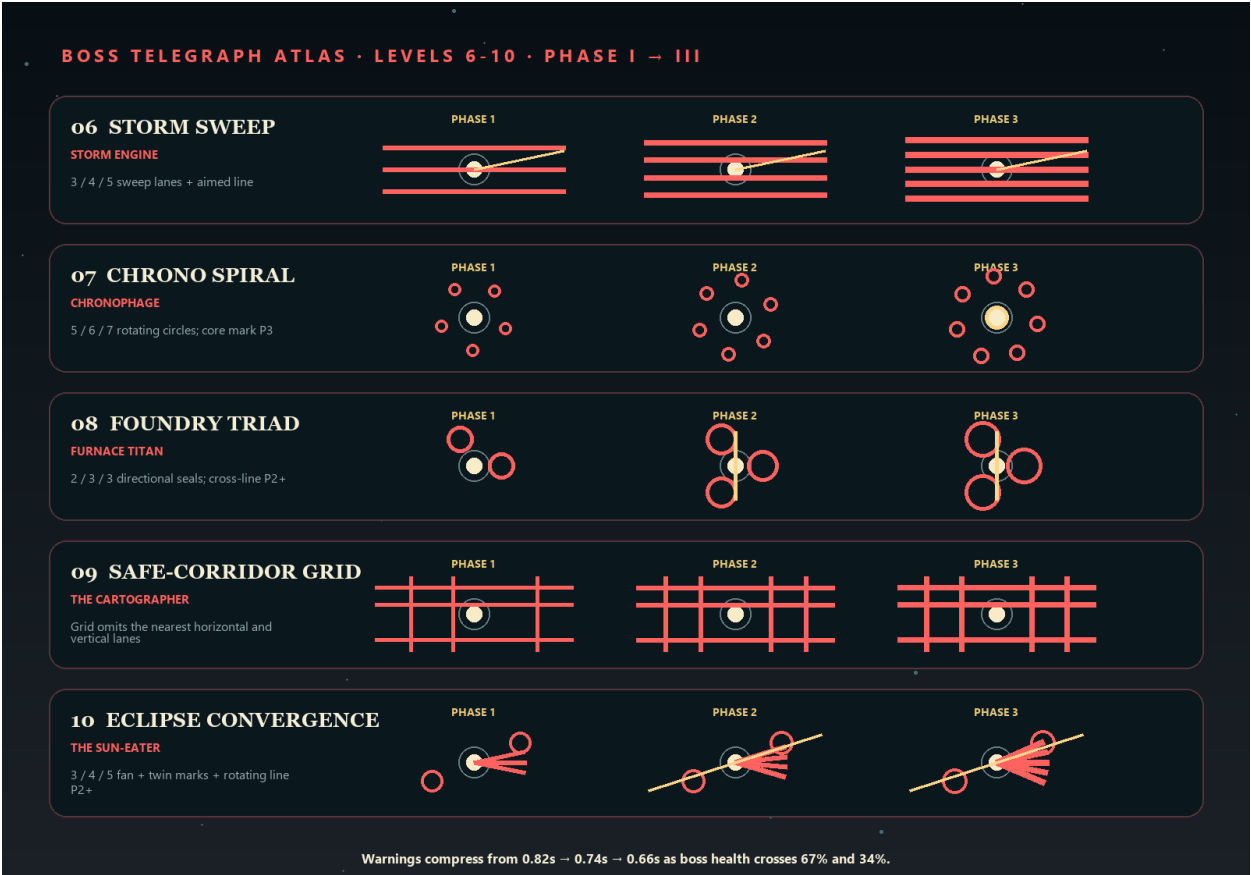
Pattern	Boss	Phase scaling	Damage
01 Line fan	Gloam Stag	2 → 3 → 4 aimed lines; width 45 → 55	x1.15
02 Orbit ring	Mire Cantor	4 → 5 → 6 circles; radius 62 → 70	x1.00
03 Cross lanes	Railjaw Prime	2 → 2 → 4 lanes; width 41 → 51	x1.12
04 Mirror marks	Mirror Matron	Twin marks; aimed line from Phase II	Circle x1.00 • line x1.12
05 Spiral cluster	Tide Apostle	3 → 4 → 5 circles; stagger 0.06s	x1.08

Sources: balance.ts BOSS_PATTERN_COUNT / bossPatternForLevel • GameCanvas.tsx bossAttack • animation.ts

BOSS SYSTEMS 02

Boss Attack Pattern Atlas • Levels 6–10

Each sector now owns a distinct three-phase telegraph signature.



Boss phase thresholds: Phase II at <=67% HP; Phase III at <=34% HP. Warning time: 0.82s / 0.74s / 0.66s.

Pattern	Boss	Phase scaling	Damage
06 Storm sweep	Storm Engine	3 → 4 → 5 sweep lanes + aimed line	Sweep x1.00 · aim x1.08
07 Chrono spiral	Chronophage	5 → 6 → 7 rotating circles; core mark P3	Core P3 x1.08
08 Foundry triad	Furnace Titan	2 → 3 → 3 seals; perpendicular line P2+	Seal x1.06 · line x1.00
09 Safe-corridor grid	Cartographer	Vertical + horizontal grid omits nearest lane in each axis	x1.00
10 Eclipse convergence	Sun-Eater	3 → 4 → 5 fan + twin marks + rotating line P2+	Fan x1.06

Sources: balance.ts BOSS_PATTERN_COUNT / bossPatternForLevel · GameCanvas.tsx bossAttack · animation.ts

BOSS SYSTEMS 04

Boss Roster • Levels 1–5

Each boss enters 38 seconds before nominal level end; victory waits until the boss dies.

BOSS ROSTER • LEVELS 1–5



01 GLOAM STAG

LINE FAN

REFERENCE HP 1,910 2 / 3 / 4 lines

PHASE ATTACK GAP

2.000 → 1.729s

P1

P2

P3

ATLAS FRAME 0



02 MIRE CANTOR

ORBIT RING

REFERENCE HP 2,332 4 / 5 / 6 circles

PHASE ATTACK GAP

2.000 → 1.657s

P1

P2

P3

ATLAS FRAME 1



03 RAILJAW PRIME

CROSS LANES

REFERENCE HP 3,122 2 / 2 / 4 lanes

PHASE ATTACK GAP

2.000 → 1.598s

P1

P2

P3

ATLAS FRAME 2



04 MIRROR MATRON

MIRROR MARKS

REFERENCE HP 3,627 Twin marks; line P2+

PHASE ATTACK GAP

2.000 → 1.548s

P1

P2

P3

ATLAS FRAME 3



05 TIDE APOSTLE

SPIRAL CLUSTER

REFERENCE HP 4,532 3 / 4 / 5 circles

PHASE ATTACK GAP

1.960 → 1.505s

P1

P2

P3

ATLAS FRAME 1

HP shown is the reference-build target, not a fixed spawn value. Live HP adapts to the player's exact build.

L	Boss	Ref HP	Phase II / III HP	Chase speed P1 → P3	Pattern
1	Gloam Stag	1,910	1,280 / 649	45.2 → 57.7	Line Fan
2	Mire Cantor	2,332	1,562 / 793	46.1 → 58.8	Orbit Ring
3	Railjaw Prime	3,122	2,092 / 1,061	47.2 → 60.2	Cross Lanes
4	Mirror Matron	3,627	2,430 / 1,233	48.4 → 61.7	Mirror Marks
5	Tide Apostle	4,532	3,036 / 1,541	49.6 → 63.3	Spiral Cluster

Sources: content.ts LEVELS • balance.ts estimateBossDps / bossHealthForBuild / bossAttackRecoverySeconds • GameCanvas.tsx spawnBoss / bossAttack

BOSS SYSTEMS 05

Boss Roster • Levels 6–10

Each boss enters 38 seconds before nominal level end; victory waits until the boss dies.

BOSS ROSTER • LEVELS 6–10



06 STORM ENGINE

STORM SWEEP

REFERENCE HP 5,504 3 / 4 / 5 lanes + aim

PHASE ATTACK GAP

1.911 → 1.468s

P1

P2

P3

ATLAS FRAME 4



07 CHRONOPHAGE

CHRONO SPIRAL

REFERENCE HP 6,542 5 / 6 / 7 rings; core P3

PHASE ATTACK GAP

1.869 → 1.435s

P1

P2

P3

ATLAS FRAME 4



08 FURNACE TITAN

FOUNDRY TRIAD

REFERENCE HP 8,201 2 / 3 / 3 seals; line P2+

PHASE ATTACK GAP

1.832 → 1.406s

P1

P2

P3

ATLAS FRAME 2



09 THE CARTOGRAPHER

SAFE-CORRIDOR GRID

REFERENCE HP 9,826 Grid leaves 2 safe lanes

PHASE ATTACK GAP

1.798 → 1.381s

P1

P2

P3

ATLAS FRAME 3



10 THE SUN-EATER

ECLIPSE CONVERGENCE

REFERENCE HP 12,000 Fan + twin marks + line P2+

PHASE ATTACK GAP

1.768 → 1.358s

P1

P2

P3

ATLAS FRAME 5

HP shown is the reference-build target, not a fixed spawn value. Live HP adapts to the player's exact build.

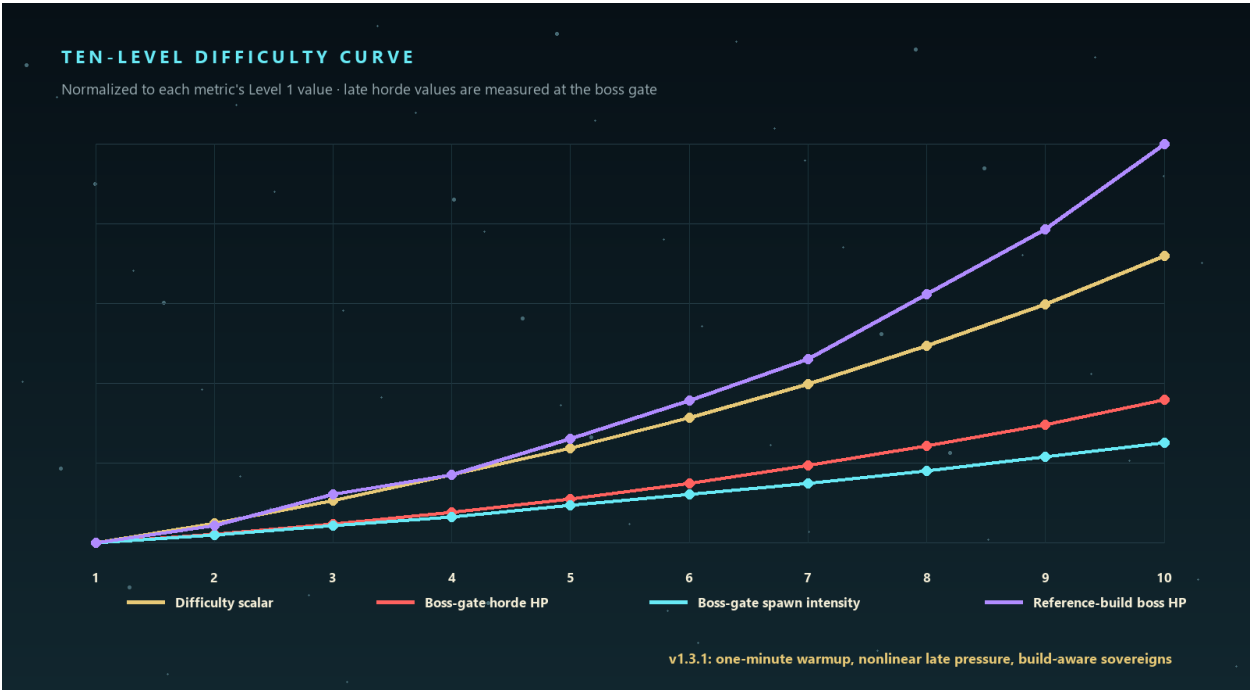
L	Boss	Ref HP	Phase II / III HP	Chase speed P1 → P3	Pattern
6	Storm Engine	5,504	3,688 / 1,871	51.0 → 65.0	Storm Sweep
7	Chronophage	6,542	4,383 / 2,224	52.5 → 67.0	Chrono Spiral
8	Furnace Titan	8,201	5,495 / 2,788	54.3 → 69.3	Foundry Triad
9	The Cartographer	9,826	6,583 / 3,341	56.2 → 71.7	Safe-Corridor Grid
10	The Sun-Eater	12,000	8,040 / 4,080	58.5 → 74.6	Eclipse Convergence

Sources: content.ts LEVELS • balance.ts estimateBossDps / bossHealthForBuild / bossAttackRecoverySeconds • GameCanvas.tsx spawnBoss / bossAttack

DIFFICULTY SYSTEMS 01

Current Difficulty Curve

The curve combines level scalars, within-run escalation, species modifiers, recurring hazards, and boss phases.



Reference-build boss HP rises from 1,910 to 12,000; late horde multipliers are measured immediately before the boss gate.

L	Level	Time	Open spawn	Gate spawn	Gate HP	Cap	Ref boss
1	First Beacon	4:00	1.13/s	8.30/s	2.67×	212	1,910
2	Glassreed Mire	4:30	1.24/s	9.11/s	2.97×	232	2,332
3	Shattered Arcade	5:00	1.38/s	10.12/s	3.33×	252	3,122
4	Prism Garden	5:00	1.51/s	11.13/s	3.74×	268	3,627
5	Drowned Docks	5:30	1.68/s	12.35/s	4.22×	287	4,532
6	Stormrail	6:00	1.80/s	13.56/s	4.75×	305	5,504
7	Hourglass Vault	6:00	1.80/s	14.78/s	5.41×	321	6,542
8	Cinder Foundry	6:30	1.80/s	16.19/s	6.09×	339	8,201
9	Void Observatory	7:00	1.80/s	17.71/s	6.83×	357	9,826
10	Crown of Dawn	7:30	1.80/s	19.23/s	7.72×	374	12,000

Sources: content.ts LEVELS / GLOBAL_DIFFICULTY_MULTIPLIER · balance.ts hordePressureAt / sectorBaselineAt / hordeActiveCap / bossTargetTtkSeconds

DIFFICULTY SYSTEMS 02

Difficulty Logic · Horde Runtime

The first minute is deliberately readable; pressure then accelerates nonlinearly to a common boss-gate endpoint.

System	Current formula	Meaning
0–60s pressure	$w = \text{smoothstep}(t/60)$; spawn $1.25 + 0.95w$; HP $.82 + .18w$; speed $.92 + .08w$; damage $.82 + .18w$	Soft opening converges to the authored baseline at 60s.
After 60s	$q = \text{clamp}((t-60)/(\text{bossAt}-60))$; spawn $2.2 + 3q + 4q^2$; HP $1 + .55q + 1.15q^2$; speed $1 + .12q$; damage $1 + .55q$	Every sector reaches 9.2 spawn / 2.7 HP / 1.12 speed / 1.55 damage at the gate.
Sector baseline	$\text{lerp}(\text{min}(\text{authored spawn}, 1.31), \text{authored spawn}, w)$; HP uses $\text{min}(\text{authored HP}, 1.5)$	High sectors no longer dump full authored pressure into second one.
Actual spawn	$\text{baselineSpawn} \times \text{pressureSpawn} \times 1.10$	Global difficulty increase is included; remaining horde runs at $\times 0.45$ during a boss.
Active cap	$\text{min}(380, 62 + 16L + \text{floor}(80p + 95p^2))$	Pre-boss caps rise from 212 to 374.
Enemy HP	$34 \times \text{baselineHP} \times \text{speciesScale} \times \text{pressureHP} \times 1.10$	Species scale ranges 0.72–1.68.
Specialist gates	Cantor 10%; Railjaw 22%; Chronowisp 30%; Cinder Guard 45%	Maskling and Shardwing are available immediately.
XP threshold	$L1=12$; then $\text{round}(16 + 8L + 2L^{1.35})$, plus $\text{round}(4(L-8)^{1.55})$ after L8	Late upgrade costs rise with the player's compounding build.
Hazards	First at 22s; then $\text{max}(5.2, 14 - 1.15 \times \text{difficulty})$	Interval contracts from 12.85s to 8.48s.

WHAT THIS FIXES

Opening pressure no longer peaks before the player has a draft. After minute one, quadratic spawn and HP growth outpace early linear rank gains, while specialist gates and the late XP tax prevent the middle and final minutes from flattening.

Sources: `balance.ts` `hordePressureAt` / `sectorBaselineAt` / `hordeActiveCap` / `eligibleEnemyPool` / `experienceToNextLevel` · `GameCanvas.tsx` `updateSpawning` / `spawnEnemy` / `hazardTimer`

DIFFICULTY SYSTEMS 03

Difficulty Logic · Build-Aware Sovereigns

Boss durability is solved from the current build, then constrained by authored floors and safety caps.

System	Current formula	Meaning
Spawn time	$\max(45, \text{duration} - 38)$	Every current boss enters with 38 nominal seconds remaining.
Target fight	$\text{TTK} = 19 + 1.1L$ seconds	Reference target rises from 20.1s in Sector 1 to 30.0s in Sector 10.
Estimated DPS	Live weapons + ranks + modules + awakenings + Trace + Astrarium; single-target uptime factors	Area and chain weapons are not credited with perfect boss uptime.
Adaptive HP	$\text{referenceDPS} \times \text{targetTTK} \times \text{clamp}(\text{DPS}/\text{referenceDPS}, 7, 6)^{.82}$	Sublinear scaling preserves phase time without fully counter-building the player.
HP bounds	$\max(\text{base} \times 1.05, \min(\text{adaptive}, \text{base} \times 16));$ $\text{base} = 1120 \times \text{enemyHealth} \times (1 + .12L) \times 1.10$	Authoring floor and hard ceiling reject degenerate inputs.
Reference DPS	95 / 110 / 140 / 155 / 185 / 215 / 245 / 295 / 340 / 400	Produces the 1,910 → 12,000 reference HP shown in the roster.
Boss damage	$18 + 3.1 \times \text{difficulty}$	Pattern family multiplies; final player damage applies x1.10 global and x0.72 incoming.
Boss chase	$(48 + 4 \times \text{difficulty}) \times 0.87 / 0.99 / 1.11$	Movement accelerates at each HP phase.
Attack recovery	$\text{clamp}((\text{targetTTK}/(9+L)) \times [1.12, 1.86], .98, 2)$	P1 → P3 ranges 2.000→1.729s at L1 and 1.768→1.358s at L10.
Warning	$0.90 - 0.08 \times \text{phase}$	0.82s → 0.74s → 0.66s.

TEN REAL SIGNATURES

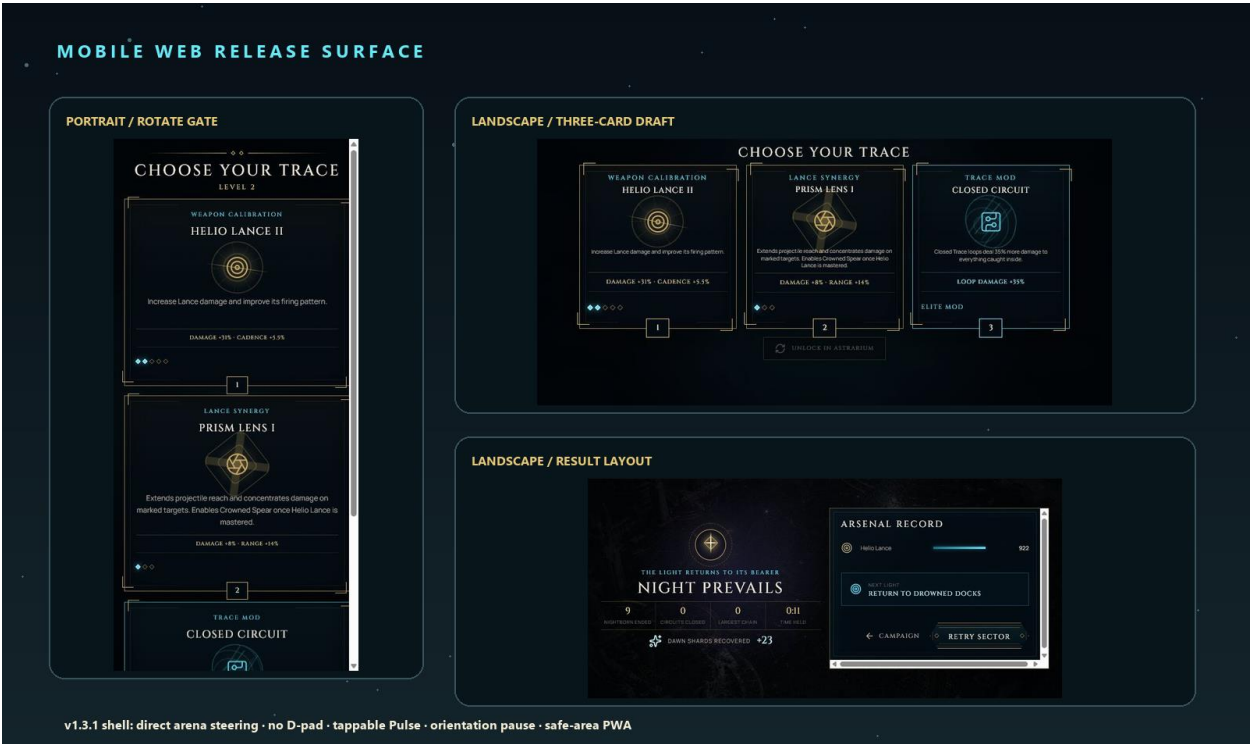
Boss pattern index = $(\text{level} - 1) \bmod 10$. The campaign now runs Line Fan, Orbit Ring, Cross Lanes, Mirror Marks, Spiral Cluster, Storm Sweep, Chrono Spiral, Foundry Triad, Safe-Corridor Grid, then Eclipse Convergence.

Sources: `balance.ts estimateBossDps / bossTargetTtkSeconds / bossAttackRecoverySeconds / bossHealthForBuild / bossPatternForLevel`
`· GameCanvas.tsx spawnBoss / bossAttack`

PLATFORM AUDIT 01

Mobile Compatibility

Verdict: mobile-web release ready over HTTPS; native App Store / Play Store packaging remains a separate release step.



Portrait combat is gated behind a rotate prompt; the landscape draft presents all three choices side by side.

Area	Status	Evidence / remaining gap
Touch input	READY	Drag anywhere in the arena to steer. No D-pad. Pulse and Pause are tappable UI controls.
Landscape combat	READY	Manifest requests landscape; combat attempts fullscreen lock and pauses behind an accessible rotate gate in portrait.
Upgrade draft	READY	All three paused choices render side by side in short landscape, with touch targets and reroll support.
Safe areas	READY	viewport-fit=cover plus consumed safe-area CSS variables protect HUD, controls, drafts, and gates.
PWA / offline	READY	Versioned service worker caches the app shell, Vite chunks, and optimized gameplay art after first online load.
Install assets	READY	Standard, maskable, and Apple touch icons; deployment-relative manifest and worker URLs.
Shareable access	READY	HTTPS release: https://jaredjos.github.io/H5-Games/nighttrace/
Native stores	PENDING	Requires wrapper identity, signing, privacy forms, store artwork, screenshots, and physical-device acceptance.

HOW TO PLAY ON A PHONE

Open the HTTPS release in a mobile browser, rotate to landscape when prompted, and use Add to Home Screen / Install app for an icon and offline play. The Windows launcher remains loopback-only and is intentionally not a phone link.

Sources: README Mobile install and offline play / Controls · manifest.webmanifest · sw.js · pwa.ts · App.tsx orientation gate · styles.css safe-area / landscape blocks · GameUI.tsx mobile-controls

APPENDIX

Implementation Notes & Source Anchors

Use this page to reconcile authored copy, visual design, and live runtime behavior.

Known design/runtime deltas

- No weapon branch-choice system is active. The unused OwnedWeapon.branch field is not an evolution feature.
- Eventide Garden, Sepulchral Orchard, Infinite Refrain, and Black Benediction add the universal awakening multipliers plus a readable final-state VFX signature. Graveglass and Harrow overlaps are presentation rather than repeated damage zones.
- Prism Lens affects lance damage, speed, and life; Gyro Crown adds blades and cooldown; Resonance Coil adds targets and initial damage without extending the 320 chain radius.
- Grav Anchor lengthens the seed and pickup attraction but does not enlarge the 92-radius bloom. Flux Mirror strengthens only the forward bolt; the reverse bolt remains 75% of rank/awakening core.
- Pulse Prime fills charge to 100%. Invulnerability is not granted by selecting it; Dawn Pulse activation provides the later hitstop event.
- Weapon figures are actual production-renderer samples, not authored diagrams. Timing matters: each frame is frozen at the weapon-specific readability beat recorded in capture-manifest.json.
- Enemy and boss animation is visual-only: procedural poses, shadows, footfalls, hover trails, normal melee choreography, and level-specific special accents do not change collision radii, movement speed, damage, or recovery.
- The original ambient and boss music layers use lifecycle-safe playback and crossfades. Reference-video audio was measured for pacing contrast only and is not included in the game.

Primary source map

Area	Source anchors
Runtime capture evidence	outputs/nighttrace-upgrade-runtime-captures/capture-manifest.json + 32 PNGs
Weapons, modules, Trace, levels	src/game/content.ts
Rank/evolution formulas and attacks	src/game/GameCanvas.tsx updateWeapons / fireWeapon
Horde and boss balance	src/game/balance.ts; src/game/GameCanvas.tsx spawning / boss paths
Support relics	src/game/balance.ts supportPickup*; src/game/pickupPresentation.ts; GameCanvas.tsx pickup paths
Persistent nodes	src/ui/data.ts; src/game/GameCanvas.tsx persistent initialization
Draft and HUD visuals	src/ui/GameUI.tsx; src/ui/Primitives.tsx; src/styles.css
Enemy and boss animation	src/game/animation.ts; src/game/GameCanvas.tsx render / triggerEnemyAttack
Music and transitions	src/game/audio.ts; public/assets/audio
Mobile shell	README.md; App.tsx; pwa.ts; index.html; manifest.webmanifest; sw.js; styles.css
Scope	NIGHTTRACE v1.6.0 working build inspected 25 July 2026. Values are implementation-derived. Weapon plates are real deterministic captures; replacement powers are fresh v1.6.0 recaptures.